

The Shutdown mode allows to achieve the lowest power consumption. The internal regulator is switched off to power down the V_{CORE} domain. The HSI48, HSIUSB48 and LSI RC-oscillators and HSE crystal oscillator are also powered down. The RTC is off.

The BOR is not available in Shutdown mode. No power voltage monitoring is possible in this mode.

SRAM and register contents are lost.

The device exits Shutdown mode upon external reset event (NRST pin), or wake-up event (any WKUP pin, configurable rising or falling edge).

3.7.5 Reset mode

During and upon exiting reset, the Schmitt triggers of I/Os are disabled so as to reduce power consumption. In addition, when the reset source is internal, the built-in pull-up resistor on NRST pin is deactivated.

3.8 Interconnect of peripherals

Several peripherals have direct connections between them. This allows autonomous communication between peripherals, saving CPU resources thus power supply consumption. In addition, these hardware connections allow fast and predictable latency.

Depending on peripherals, these interconnections can operate in Run, Sleep and Stop modes.

Table 4. Interconnect of peripherals

Interconnect source	Interconnect destination	Interconnect action	Run	Sleep	Stop
TIMx	TIMx	Timer synchronization or chaining	Y	Y	-
	ADCx	Conversion triggers	Y	Y	-
	DMA	Memory-to-memory transfer trigger	Y	Y	-
ADCx	TIM1	Timer triggered by analog watchdog	Y	Y	-
RTC	TIM16	Timer input channel from RTC events	Y	Y	-
All clock sources (internal and external)	TIM14,16,17	Clock source used as input channel for RC measurement and trimming	Y	Y	-
CSS RAM (parity error)	TIM1,16,17	Timer break	Y	Y	-
CPU (hard fault)	TIM1,16,17	Timer break	Y	-	-